

AI-AUGMENTED BACKEND DEVELOPMENT

AI BlogNest API

4. Project Planning Phase

Implementation sequence, modules, structure, testing plan, and deliverables

1. Planning Objective

Project planning converts the designed architecture into an ordered implementation workflow. The sample organizes the backend into project setup, backend development, database configuration, execution, and API testing stages. ■filecite■turn0file0■L153-L193■

2. Proposed Development Sequence

- Create the project folder and initialize the server-side Node.js project.
- Set up server.js and the models, controllers, and routes directories.
- Configure Express and application middleware.
- Configure MongoDB and Mongoose.
- Create user, post, comment, category, and related data models.
- Implement authentication and role authorization.
- Implement blog CRUD and workflow states.
- Implement comments, categories, tags, search, and media metadata.
- Integrate AI generation and summarization services.
- Apply validation, sanitization, rate limiting, and logging.
- Test all endpoints with Postman or Thunder Client.

3. Module Planning

- Authentication module: registration, login, password hashing, JWT.
- Authorization module: role matrix and protected routes.
- Blog module: CRUD, lifecycle states, likes, and ownership.
- Taxonomy module: categories and tags.
- Comment module: creation and moderation.
- AI module: generation and summarization.
- Database module: MongoDB/Mongoose connection and schemas.
- Security module: validation, sanitization, rate limiting, and centralized errors.

4. Project Structure

The sample begins the server structure with `server.js` and separate `models`, `controllers`, and `routes` folders. This separation should be retained so that data definitions, request handling, and route declarations do not become tightly coupled. ■filecite■turn0file0■L160-L169■

5. Testing Plan

- Test successful and failed registration.
- Test successful and failed login.
- Verify JWT-protected routes.
- Verify each role's permitted and forbidden actions.
- Test blog creation, retrieval, modification, and deletion.
- Test comments and moderation.
- Test AI generation and summarization responses.
- Test invalid payloads, unauthorized requests, and malformed parameters.
- Test database connectivity and error handling.

6. Deliverables

- Configured Node.js backend project.
- MongoDB database and Mongoose models.
- REST API routes and controllers.
- Authentication and authorization middleware.
- AI generation and summarization services.
- Postman/Thunder Client test collection.
- Project documentation and demonstration material.